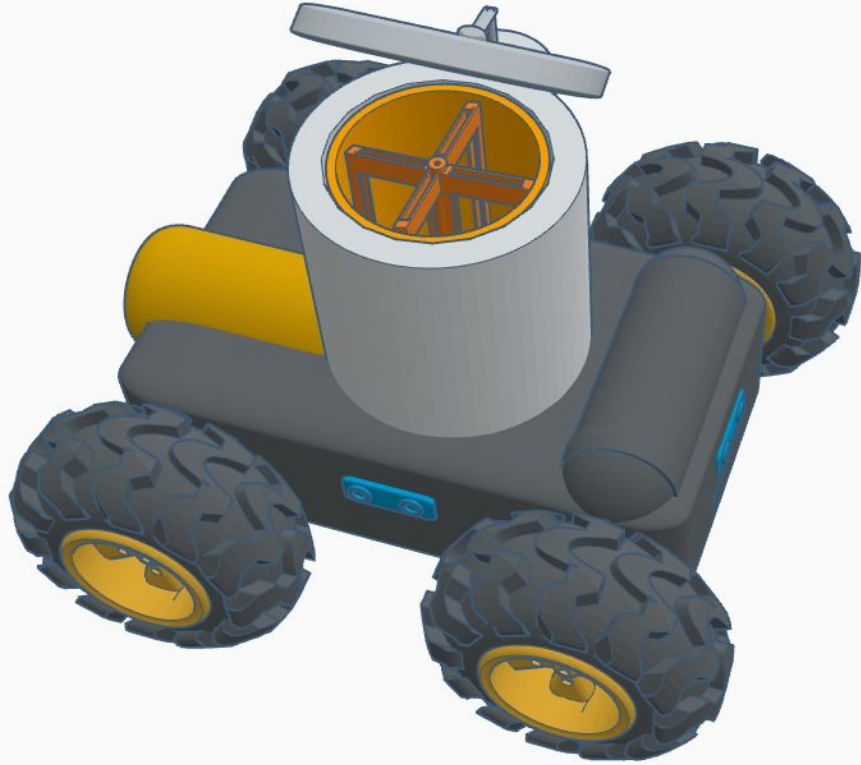


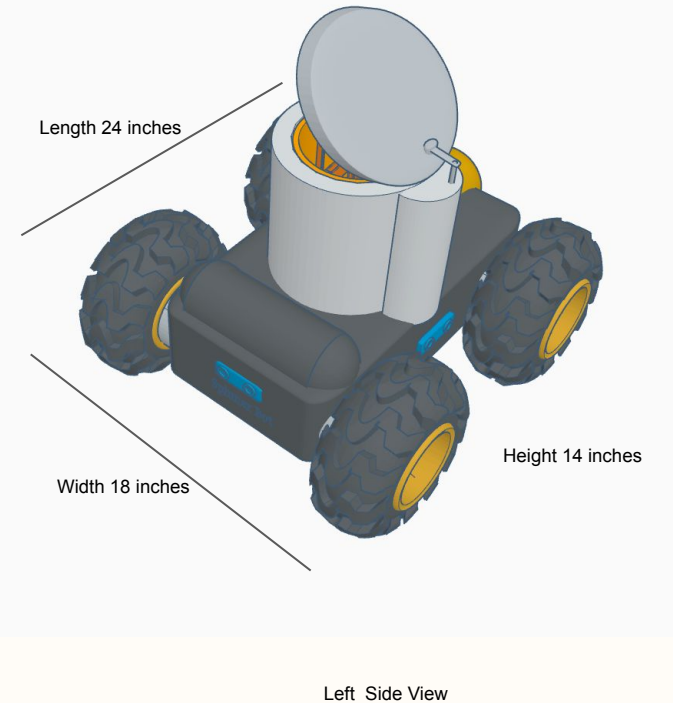
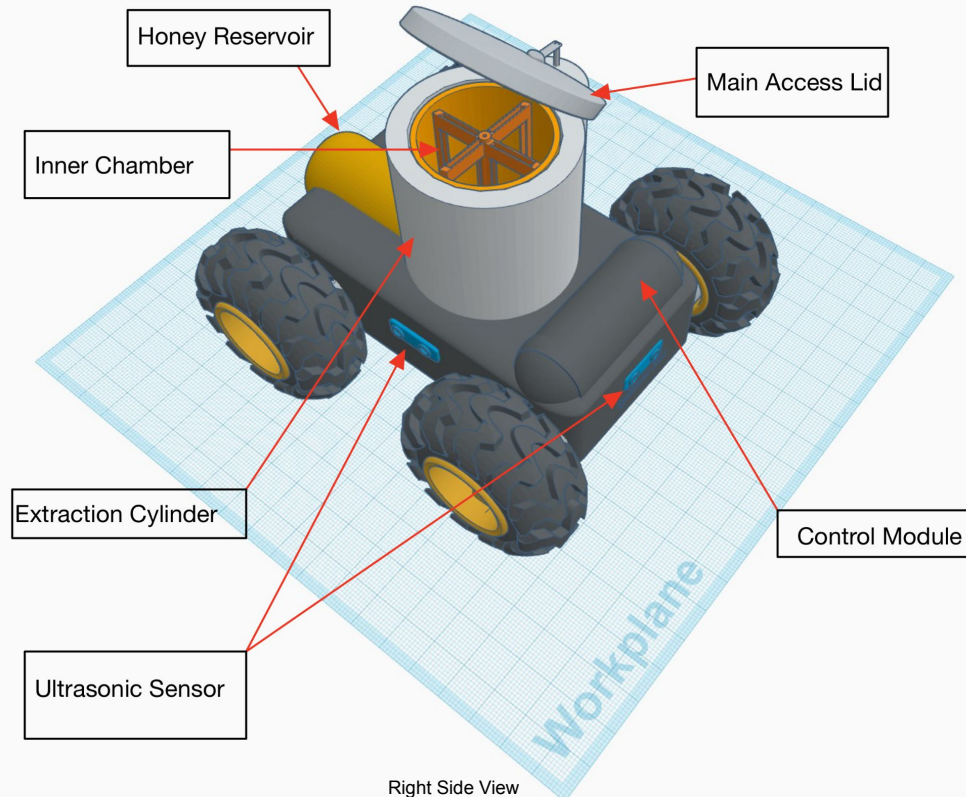


The Spinner Bot

By Jason henry

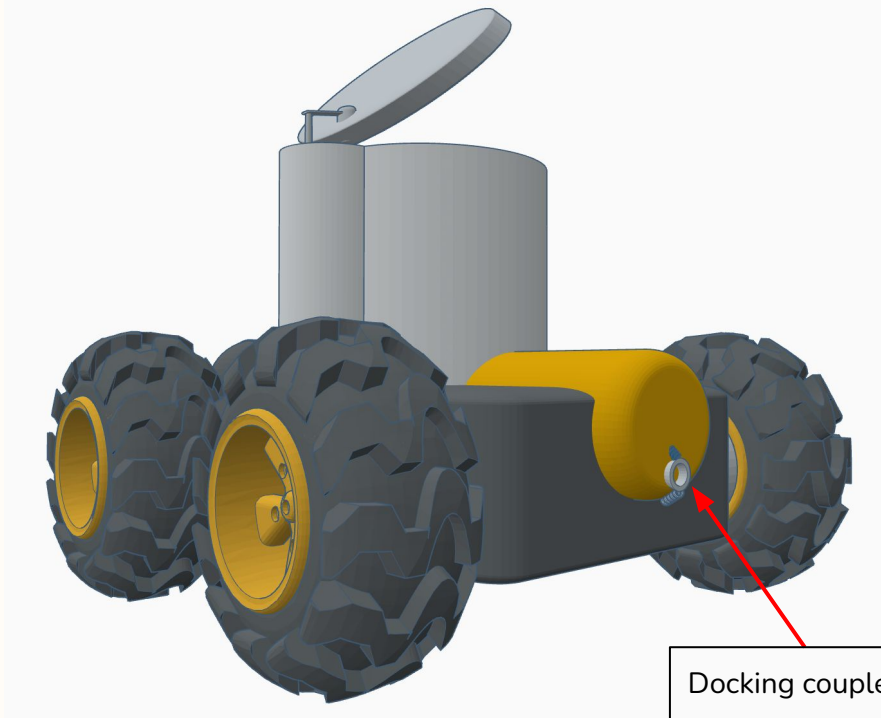
MECHANICAL DESIGN

Spinner Bot

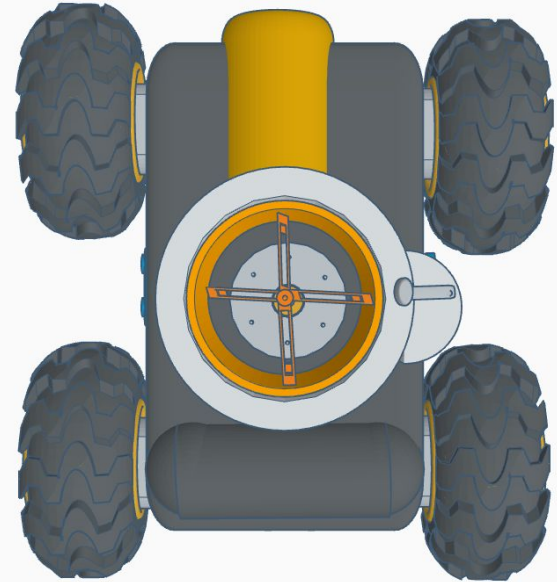


MECHANICAL DESIGN

Spinner Bot



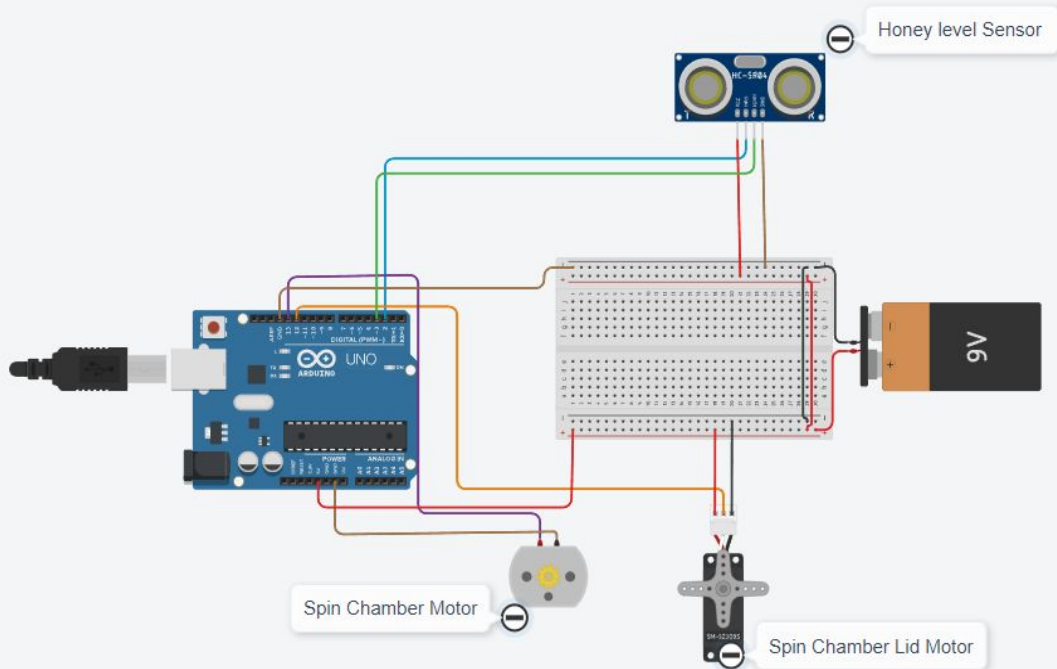
Back View



Top View

CIRCUIT DESIGN

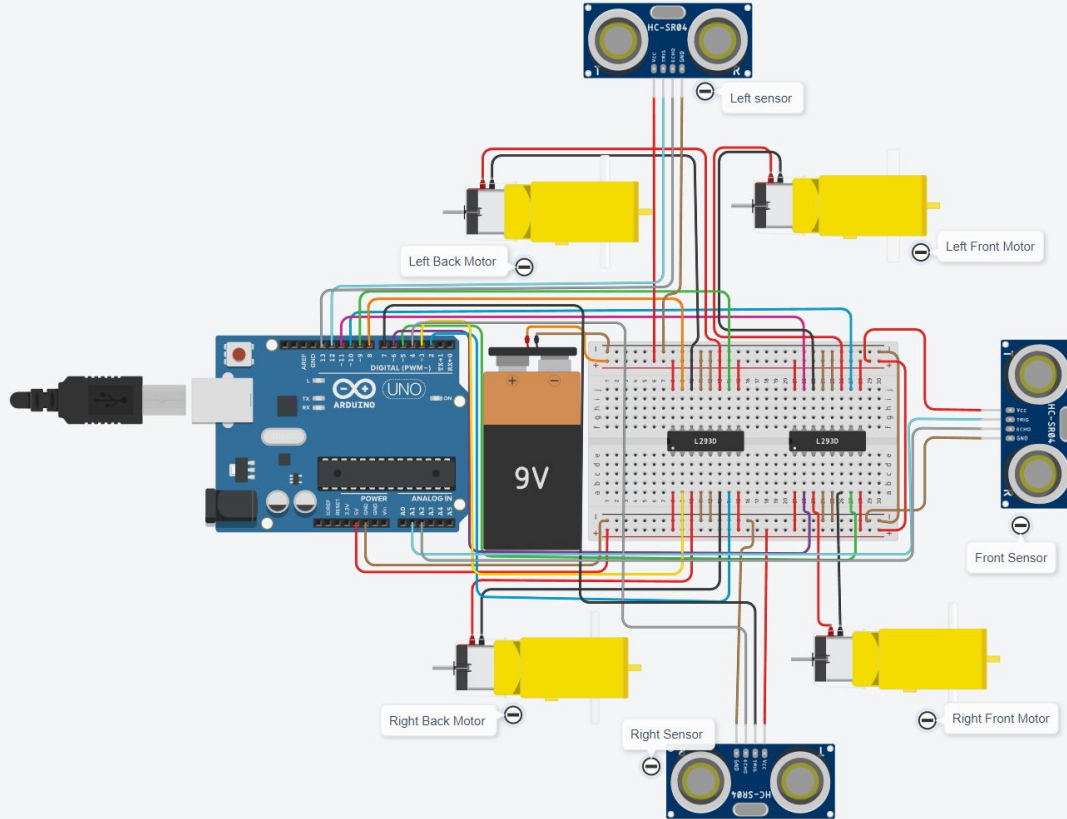
Spinner Bot-Extraction Chamber



Enables automation of honey extraction utilizing one ultrasonic distance sensor for honey level readings, one servo motor for opening and closing of chamber door, and one DC motor for spinning extraction

CIRCUIT DESIGN

Spinner Bot Movement



Provides movement capabilities for the spinner bot, utilizing three ultrasonic distance sensor in conjunction with four DC motors

ARDUINO CODE

Spinner Bot-Extraction Chamber

```
Text
1 #include <Servo.h>
2
3 const int trigPin = 2;
4 const int echoPin = 1;
5 const int spinMotorPin = 13;
6 const int servoPin = 12;
7
8 const int HONEY_LEVEL_HIGH_THRESHOLD = 5;
9 const int HONEY_LEVEL_LOW_THRESHOLD = 15;
10
11 const unsigned long SPIN_CYCLE_DURATION = 180000; // 3 minutes
12 bool isSpinning = false;
13 unsigned long spinCycleStartTime = 0;
14 unsigned long lastPrintTime = 0;
15
16 Servo doorServo;
17
18 // Servo positions
19 const int DOOR_OPEN_POSITION = 90;
20 const int DOOR_CLOSE_POSITION = 0;
21
22 void setup() {
23   Serial.begin(9600);
24
25   pinMode(trigPin, OUTPUT);
26   pinMode(echoPin, INPUT);
27   pinMode(spinMotorPin, OUTPUT); // Spinner motor
28
29   doorServo.attach(servoPin); // Servo for chamber door
30   doorServo.write(DOOR_CLOSE_POSITION); // Start with the door closed
31 }
32
33 long measureDistance() {
34   digitalWrite(trigPin, LOW);
35   delayMicroseconds(2);
36   digitalWrite(trigPin, HIGH);
37   delayMicroseconds(10);
38   digitalWrite(trigPin, LOW);
39
40   long duration = pulseIn(echoPin, HIGH, 25000); // 25ms timeout
41   long distance = (duration / 2) / 29.1; // Convert to distance in cm
42
43   return distance;
44 }
45
46 void openDoor() {
47   doorServo.write(DOOR_OPEN_POSITION);
48   delay(500);
49 }
50
51 void closeDoor() {
52   doorServo.write(DOOR_CLOSE_POSITION);
53   delay(500);
54 }
55
56 void startSpinCycle() {
57   if (!isSpinning) {
58     digitalWrite(spinMotorPin, HIGH); // Start the spinner motor
59     isSpinning = true;
60     spinCycleStartTime = millis();
61     Serial.println("Spinning Started");
62   } else {
63     Serial.println("Already Spinning");
64   }
65 }
66
67 void stopSpinCycle() {
68   if (isSpinning) { // Only stop if it's currently spinning
69     digitalWrite(spinMotorPin, LOW); // Stop the spinner motor
70     isSpinning = false;
71     Serial.println("Spinning Stopped");
72   } else {
73     Serial.println("Not Spinning");
74   }
75 }
76
77 void loop() {
78   // Check if spin cycle needs to be stopped after 3 minutes
79   if (isSpinning && (millis() - spinCycleStartTime) >= SPIN_CYCLE_DURATION) {
80     stopSpinCycle();
81   }
82
83   // Check for Serial commands
84   if (Serial.available()) {
85     char command = Serial.read();
86     switch (command) {
87       case 'c':
88         Serial.println("DockingComplete");
89         break;
90       case 'd':
91         Serial.println("DockingReady");
92         break;
93       case 'f':
94         Serial.println("Full");
95         break;
96       case 'h':
97         Serial.println("HoneyLevelHigh");
98         break;
99       case 'l':
100        Serial.println("HoneyLevelLow");
101        break;
102       case 'n':
103        Serial.println("SpinCycleOn");
104        startSpinCycle();
105        break;
106       case 'x':
107        Serial.println("SpinCycleOff");
108        stopSpinCycle();
109        break;
110       case 'p':
111        Serial.println("ValueOpen");
112        break;
113       case 'u':
114        Serial.println("ValveClosed");
115        break;
116       case 'z':
117        Serial.println("Hive ID Ready");
118        break;
119       case 'a':
120        Serial.println("ChamberOpen");
121        openDoor();
122        break;
123       case 'q':
124        Serial.println("ChamberClose");
125        closeDoor();
126        break;
127       default:
128        Serial.println("Unknown command");
129        break;
130     }
131   }
132 }
133
134 if (isSpinning && (millis() - lastPrintTime) >= 5000) { // print every 5 seconds
135   Serial.print("Spin Cycle Time: ");
136   Serial.println(millis() - spinCycleStartTime);
137   lastPrintTime = millis();
138 }
139
140 delay(200);
141
142 // Check honey level
143 long distance = measureDistance();
144 Serial.print("Measured Distance: ");
145 Serial.println(distance);
146
147 if (distance <= HONEY_LEVEL_HIGH_THRESHOLD) {
148   Serial.println("HoneyLevelHigh");
149 } else if (distance >= HONEY_LEVEL_LOW_THRESHOLD) {
150   Serial.println("HoneyLevelLow");
151 }
152 }
```

ARDUINO CODE

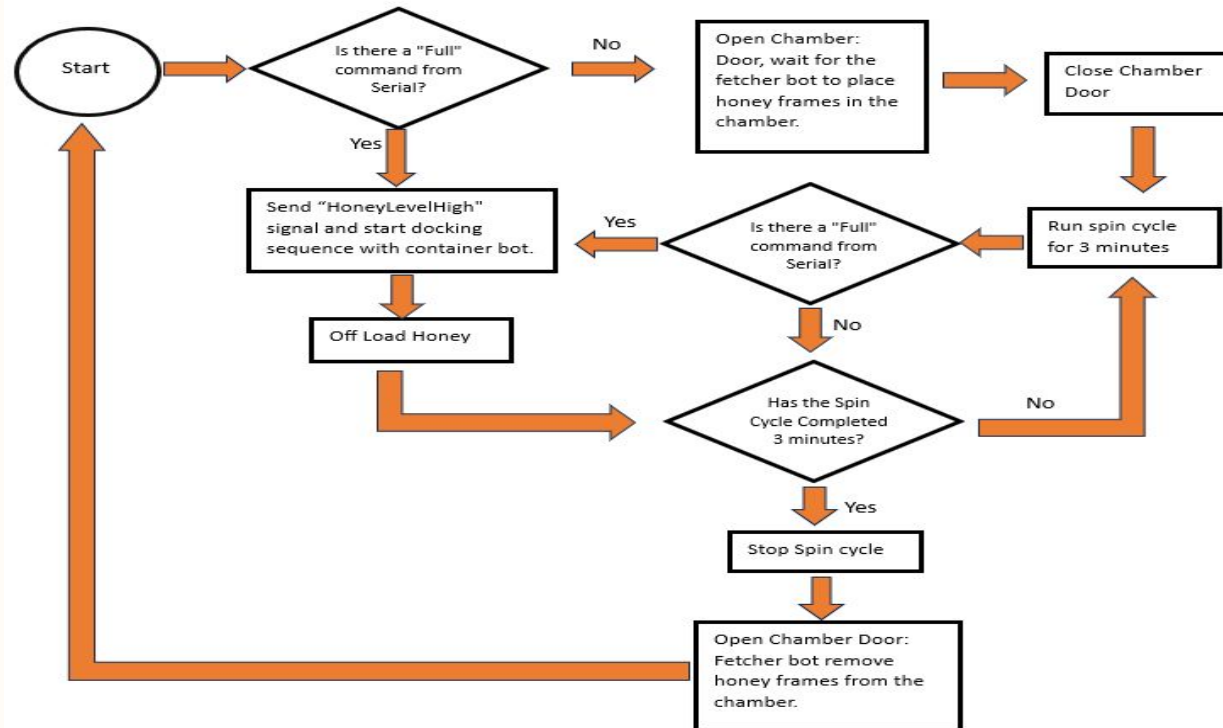
Spinner Bot Movement

```
Text 2 (Arduino Uno R3) Text 2 (Arduino Uno R3)
1 // Define pins for the motors
2 const int leftFrontForward = 10;
3 const int leftFrontBackward = 11;
4 const int leftBackForward = 9;
5 const int leftBackBackward = 8;
6 const int rightFrontForward = 6;
7 const int rightFrontBackward = 5;
8 const int rightBackForward = 3;
9 const int rightBackBackward = 2;
10
11 // Define pins for the ultrasonic sensors
12 const int leftTrigPin = 12;
13 const int leftEchoPin = 13;
14 const int frontTrigPin = A1;
15 const int frontEchoPin = A2;
16 const int rightTrigPin = 7;
17 const int rightEchoPin = 4;
18
19 const int distanceThreshold = 15; // in inches
20
21 bool robotActive = true;
22
23 void setup() {
24   Serial.begin(9600);
25
26   // Set motor pins as outputs
27   pinMode(leftFrontForward, OUTPUT);
28   pinMode(leftFrontBackward, OUTPUT);
29   pinMode(leftBackForward, OUTPUT);
30   pinMode(leftBackBackward, OUTPUT);
31   pinMode(rightFrontForward, OUTPUT);
32   pinMode(rightFrontBackward, OUTPUT);
33   pinMode(rightBackForward, OUTPUT);
34   pinMode(rightBackBackward, OUTPUT);
35
36   // Set ultrasonic sensor pins
37   pinMode(leftTrigPin, OUTPUT);
38   pinMode(leftEchoPin, INPUT);
39   pinMode(frontTrigPin, OUTPUT);
40   pinMode(frontEchoPin, INPUT);
41   pinMode(rightTrigPin, OUTPUT);
42   pinMode(rightEchoPin, INPUT);
43 }
44
45 long measureDistance(int trigPin, int echoPin) {
46   digitalWrite(trigPin, LOW);
47   delayMicroseconds(2);
48   digitalWrite(trigPin, HIGH);
49   delayMicroseconds(10);
50   digitalWrite(trigPin, LOW);
51
52   long duration = pulseIn(echoPin, HIGH);
53   long distance = (duration/2) / 29.1; // Convert to distance in cm
54   distance = distance * 0.393701; // Convert cm to inches
55
56   return distance;
57 }
58
59 void forward() {
60   digitalWrite(leftFrontForward, HIGH);
61   digitalWrite(leftFrontBackward, LOW);
62   digitalWrite(leftBackForward, HIGH);
63   digitalWrite(leftBackBackward, LOW);
64   digitalWrite(rightFrontForward, HIGH);
65   digitalWrite(rightFrontBackward, LOW);
66   digitalWrite(rightBackForward, HIGH);
67   digitalWrite(rightBackBackward, LOW);
68 }
69
70 void reverse() {
71   digitalWrite(leftFrontForward, LOW);
72   digitalWrite(leftFrontBackward, HIGH);
73   digitalWrite(leftBackForward, LOW);
74   digitalWrite(leftBackBackward, HIGH);
75   digitalWrite(rightFrontForward, LOW);
76   digitalWrite(rightFrontBackward, HIGH);
77   digitalWrite(rightBackForward, LOW);
78   digitalWrite(rightBackBackward, HIGH);
79 }
80
81 void turnLeft() {
82   digitalWrite(leftFrontForward, LOW);
83   digitalWrite(leftFrontBackward, HIGH);
84   digitalWrite(leftBackForward, LOW);
85   digitalWrite(leftBackBackward, HIGH);
86   digitalWrite(rightFrontForward, HIGH);
87   digitalWrite(rightFrontBackward, LOW);
88   digitalWrite(rightBackForward, HIGH);
89   digitalWrite(rightBackBackward, LOW);
90 }
91
92 void turnRight() {
93   digitalWrite(leftFrontForward, HIGH);
94   digitalWrite(leftFrontBackward, LOW);
95   digitalWrite(leftBackForward, HIGH);
96   digitalWrite(leftBackBackward, LOW);
97   digitalWrite(rightFrontForward, LOW);
98   digitalWrite(rightFrontBackward, HIGH);
99   digitalWrite(rightBackForward, LOW);
100  digitalWrite(rightBackBackward, HIGH);
101 }
102
103 void stopRobot() {
104   digitalWrite(leftFrontForward, LOW);
105   digitalWrite(leftFrontBackward, LOW);
106   digitalWrite(leftBackForward, LOW);
107   digitalWrite(leftBackBackward, LOW);
108   digitalWrite(rightFrontForward, LOW);
109   digitalWrite(rightFrontBackward, LOW);
110   digitalWrite(rightBackForward, LOW);
111   digitalWrite(rightBackBackward, LOW);
112 }
113
114 void loop() {
115   long leftDistance = measureDistance(leftTrigPin, leftEchoPin);
116   long frontDistance = measureDistance(frontTrigPin, frontEchoPin);
117   long rightDistance = measureDistance(rightTrigPin, rightEchoPin);
118
119   if (robotActive) {
120     if (frontDistance > distanceThreshold) {
121       forward();
122     } else {
123       stopRobot();
124     }
125     if (leftDistance > rightDistance) {
126       reverse();
127       delay(500);
128       turnLeft();
129       delay(500);
130     } else {
131       reverse();
132       delay(500);
133       turnRight();
134       delay(500);
135     }
136   } else {
137     stopRobot();
138   }
139
140   if (Serial.available()) {
141     String command = Serial.readStringUntil('\n');
142     command.trim();
143
144     if (command == "x") {
145       Serial.println("Hive, ID, Ready");
146       robotActive = false;
147     } else if (command == "k") {
148       Serial.println("Honey_Extraction_Complete");
149       robotActive = true;
150     } else if (command == "d") {
151       Serial.println("DockingReady");
152     }
153   }
154
155   delay(200);
156 }
157
```

ARDUINO CODE

Activity Diagram

Spinner Bot-Extraction



ARDUINO DEMONSTRATION

Spinner Bot

<https://drive.google.com/file/d/1aUYzDUXG6DZGhiKRYHL8i2-rjtVTG-40/view?usp=sharing>

The above link contains the Spinner Bot presentation video